AI-Powered Enterprise E-Commerce Platform

Business Requirements Document (BRD)

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1. Introduction

Electronic commerce (E-Commerce) has become one of the largest industries in the world. Millions of customers purchase products online every day.

However, modern businesses face many challenges such as:

- increasing competition,
- changing customer expectations,
- fraudulent activities,
- inventory management issues,
- and difficulty in understanding customer behavior.

Traditional e-commerce systems usually provide only basic functionalities and often lack intelligent decision-making capabilities.

This project proposes an AI-powered enterprise e-commerce platform that combines traditional software engineering with Artificial Intelligence technologies to improve customer experience and business efficiency.

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2. Purpose of the Document

The purpose of this document is to define the business needs and objectives of the proposed system.

This document helps:

- understand the business problems,
- identify business expectations,
- define project goals,
- establish project scope,
- and provide guidance for future development activities.

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3. Business Background

The rapid growth of online shopping has created new opportunities and challenges for businesses.

Customers now expect:

- personalized recommendations,
- fast services,
- secure transactions,
- intelligent support systems,
- and seamless shopping experiences.

At the same time, businesses need:

- better analytics,
- demand forecasting,
- fraud prevention,
- and efficient inventory management.

Artificial Intelligence technologies have become an important solution for addressing these challenges.

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4. Existing Business Problems

4.1 Poor Customer Personalization

Many e-commerce platforms provide the same shopping experience for every customer.

Problems:

- customers cannot easily discover products,
- lower customer engagement,
- reduced sales opportunities.

4.2 Fraudulent Activities

Online businesses frequently experience:

- fake orders,
- stolen payment methods,
- suspicious customer activities.

These activities can cause financial losses and reduce customer trust.

4.3 Inventory Management Issues

Businesses often struggle to determine:

- how much stock to maintain,
- which products will become popular,
- when products should be reordered.

This can result in:

- overstocking,
- understocking,
- increased operational costs.

4.4 Lack of Business Intelligence

Managers often lack sufficient information about:

- customer behavior,
- sales trends,
- product performance,
- future business opportunities.

4.5 Increasing Competition

The e-commerce industry is highly competitive.

Businesses must continuously improve:

- customer experience,
- service quality,
- and operational efficiency.

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5. Business Goals

The proposed system aims to achieve the following goals.

Goal 1

Improve customer satisfaction.

Goal 2

Increase sales revenue.

Goal 3

Reduce fraudulent activities.

Goal 4

Improve inventory management.

Goal 5

Provide intelligent business insights.

Goal 6

Support data-driven decision making.

Goal 7

Build a scalable enterprise platform.

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6. Business Objectives

The following measurable objectives have been identified.

Objective 1

Increase customer engagement through personalized recommendations.

Objective 2

Reduce inventory-related losses.

Objective 3

Improve sales forecasting accuracy.

Objective 4

Improve customer retention.

Objective 5

Provide real-time business analytics.

Objective 6

Reduce manual business operations through automation.

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7. Business Requirements

BR-01 Customer Management

The system shall allow customers to:

- create accounts,
- login securely,
- manage personal information,
- view order history.

BR-02 Product Management

The system shall allow administrators to:

- add products,
- update products,
- delete products,
- manage categories.

BR-03 Inventory Management

The system shall:

- monitor stock levels,
- generate inventory reports,
- support demand forecasting.

BR-04 Order Management

The system shall:

- process orders,
- manage order status,
- maintain transaction history.

BR-05 Recommendation System

The system shall:

- analyze customer behavior,
- recommend relevant products,
- improve personalization.

BR-06 Fraud Detection

The system shall:

- monitor suspicious activities,
- identify unusual transactions,
- generate fraud alerts.

BR-07 Business Analytics

The system shall provide:

- sales reports,
- customer reports,
- product performance reports,
- forecasting reports.

BR-08 AI Assistant

The system shall provide an AI assistant capable of:

- answering customer questions,
- assisting administrators,
- generating intelligent insights.

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8. Business Rules

Business rules define the policies that govern system operations.

Rule 1

Each customer must have a unique account.

Rule 2

Products cannot be purchased if stock is unavailable.

Rule 3

Administrators can only access management modules after authentication.

Rule 4

Fraud alerts must be generated for suspicious activities.

Rule 5

Only authorized users can access sensitive information.

Rule 6

Inventory updates must occur after successful purchases.

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9. Key Performance Indicators (KPIs)

The success of the system can be measured using the following indicators.

Customer KPIs

- Customer Satisfaction Score
- Customer Retention Rate
- Recommendation Click Rate

Business KPIs

- Sales Growth
- Revenue Growth
- Inventory Turnover Ratio

AI KPIs

- Recommendation Accuracy
- Fraud Detection Accuracy
- Forecasting Accuracy

Technical KPIs

- System Availability
- Response Time
- Number of Concurrent Users Supported

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10. Assumptions

The following assumptions have been made.

1. Customers have internet access.
2. Businesses are willing to adopt AI technologies.
3. Sufficient data will be available for AI model training.
4. Users possess basic digital literacy.

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11. Constraints

The project may face several constraints.

Technical Constraints

- Limited computational resources.
- Availability of datasets.
- Integration complexity.

Financial Constraints

- Limited project budget.
- Infrastructure costs.

Time Constraints

- Limited development time.
- Learning new technologies.

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12. Risks

Risk 1

Insufficient data for AI models.

Risk 2

Integration challenges between Java and Python services.

Risk 3

Security vulnerabilities.

Risk 4

Performance issues under heavy traffic.

Risk 5

Requirement changes during development.

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13. Expected Business Value

The proposed system is expected to provide significant value.

Value for Customers

- Better shopping experience.
- Personalized recommendations.
- Faster product discovery.

Value for Business Owners

- Increased profits.
- Better decision making.
- Reduced operational costs.
- Improved customer retention.

Value for Developers

- Exposure to enterprise architecture.
- Experience with AI integration.
- Experience with microservices and cloud technologies.

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14. Success Criteria

The project will be considered successful if:

1. Customer satisfaction improves.
2. Sales increase.
3. Fraud detection performs effectively.
4. Forecasting accuracy reaches acceptable levels.
5. The system remains scalable and maintainable.

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15. Conclusion

The AI-Powered Enterprise E-Commerce Platform aims to address major business challenges faced by modern e-commerce organizations.

By integrating Artificial Intelligence with enterprise software engineering, the system will provide:

- intelligent recommendations,
- fraud detection,
- forecasting capabilities,
- improved customer experiences,
- and data-driven decision support.

This project will serve as a modern enterprise solution capable of supporting future business growth.

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